EXHIBIT B: ESCAPED ALBERT ARRAY (E-AA) WORM ATTESTATION

Proof of Compliance with 17 CFR § 240.17a-4(f) Electronic Storage Standards

1. Mathematical Proof of Non-Invertibility

The Escaped Albert Array (E-AA) attribution pipeline maps session state data $S_{session}$ and hardware provenance vectors P_{puf} into an immutable, non-rewriteable hash payload $V(I_{author})$ defined by the recursive folding function:

$$V(I_{author}) = \mathbb{H}\Big(S_{session} \;\|\; P_{puf} \;\|\; \bigotimes_{i=1}^{n} \text{NovaFold}(\pi_i)\Big) \tag{1}$$

Where:

- $\mathbb{H}$ represents a collision-resistant SHA3-512 cryptographic hash function.
- $\text{NovaFold}(\pi_i)$ is an Incrementally Verifiable Computation (IVC) proof cycle accumulated over epoch i.
- P_{puf} is the physical unclonable optical phase-noise entropy key.

Given the properties of $\mathbb{H}$, deriving $S_{session}$ from $V(I_{author})$ requires $O(2^{512})$ operations, satisfying non-reversibility while enforcing strict WORM provenance tracking.

2. Silicon Phase-Change Material (PCM) Hardware Lock Attestation

State transition records are committed to physical silicon optics on the TIP SmartNIC using Phase-Change Materials (Ge2Sb2Te5). The transition from amorphous to crystalline state alters the optical refractive index irreversibly:

$$\Delta n_{pcm} = n_{crystalline} - n_{amorphous} \approx 1.5 + 0.7i \tag{2}$$

This physical state change forces a permanent 180° optical phase shift on target waveguides, providing an immutable, non-erasable, physical WORM record at the silicon hardware layer that complies with SEC Rule 17a-4(f) requirements.

3. Attestation Signature

The undersigned certifies that the mathematical specifications and physical hardware mechanics detailed herein enforce immutable, non-erasable, and non-rewriteable record retention.

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